CAPSTONE

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Little Guide I Made To have Some Kind Of Checklist For The Year.
There Is No Particular Order Of Things

-Dominic

Documentation

Things to do first (0)

Meet with Jay Johns and choose a clear stack
Develop initial scope what we want accomplished

Initial Project Proposal (1)

Title Page
Project title, team members, sponsor, and faculty advisor.
Abstract/Executive Summary
A brief overview of the project, including objectives, scope, and expected outcomes.
Problem Statement
A clear definition of the problem the project aims to solve.
Project Goals and Objectives
Specific, measurable goals and objectives for the project.
Scope
Define what is included in the project and what is outside the scope.
Methodology
Outline the approach, tools, and technologies you will use.
Timeline
High-level project timeline, including major milestones.
Risk Analysis
Identify potential risks and how you plan to mitigate them.

Requirement Specifications (2)

Functional Requirements
Detailed list of what the system should do (e.g., "The system shall track eye movements using a standard webcam").
Non-Functional Requirements
Performance, security, usability, and other quality attributes.
Use Cases/User Stories
Specific scenarios detailing how different users will interact with the system.
User Interface Requirements
Specifications for the user interface, including wireframes if applicable.

Design Documentation (3)

System Architecture Document
Detailed description of the system's architecture, including the interaction between components.
Data Design
Database schema, data storage structures, and data flow.
Interface Design
Design of the user interface, including mockups or prototypes.
Algorithm Design
Description of key algorithms, particularly for eye-tracking and data analysis.

Implementation Plan (4)

Development Process
Outline the development approach (e.g., Agile, Waterfall).
Version Control
Details on how version control will be managed (e.g., using Git).
Coding Standards
Guidelines for coding, including naming conventions, code organization, and documentation.

Testing Documentation (5)

Test Plan
Overall strategy for testing, including unit, integration, system, and user acceptance testing.
Test Cases
Specific test cases with expected outcomes.
Test Results
Documentation of the results of testing, including any issues found and resolved.

Deployment and Maintenance Documentation (6)

Deployment Plan
Steps required to deploy the application, including environment setup.
User Manual
Instructions for end-users on how to use the software.
Maintenance Plan
Guidelines for future updates, bug fixes, and improvements.

Final Report (7)

Summary of the Project
Recap of the project objectives and outcomes.
Challenges Faced

Description of any major challenges and how they were addressed.

Future Work

Suggestions for future improvements or extensions of the project.

References

List of references used in the project.

Diagrams & Models

System Architecture Diagram (1)

A high-level diagram showing the overall structure of the system, including major components and their interactions

Data Flow Diagram (2)

Diagrams showing how data moves through the system, from input to processing to output

Entity-Relationship Diagram (3)

A diagram that represents the database structure, including entities, relationships, and attributes

Use Case Diagrams (4)

Diagrams showing the different use cases and how various users interact with the system

Sequence Diagrams (5)

Diagrams illustrating the sequence of interactions between different components of the system for specific use cases.

Class Diagrams (6)

Diagrams showing the structure of the code, including classes, attributes, methods, and relationships between classes.

Component Diagrams (7)

Diagrams detailing the structure of specific components of the system, showing their internal workings and how they interact with other components.

Deployment Diagram (8)

A diagram showing how the software will be deployed in the environment, including servers, databases, and network configurations.

Data Models (9)

Models representing the structure and organization of data within the system

Process Models (10)

Models that describe the processes within the system, often using BPMN (Business Process Model and Notation) or similar tools.